Pennysia: An Open Liquidity Network

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Abstract

Pennysia is an open liquidity network, a single exchange layer where every source of liquidity, wherever it sits, competes for the user's order. Decentralized exchanges, aggregators, intent solvers, market makers, perpetual and prediction protocols, vaults, fiat ramps, and licensed venues all answer the same auction. The best executable price wins. The user signs once. The network routes, executes, and verifies the fill. For builders, a venue connects once and competes for every order from day one.

Pennysia aims to combine the performance, liquidity access, and usability of centralized exchanges (CEXs) with the self-custody of decentralized exchanges (DEXs). The complexity of wallet management, gas payments, and chain routing is abstracted away, while users retain their choice of wallet and chain. Fiat enters and exits on card and bank rails. Liquidity is organized into two lanes: a **permissionless lane** of open venues, onchain and offchain, open to everyone, and a **regulated lane** of gated liquidity (centralized market makers, licensed venues, compliant pools, fiat ramps) unlocked by optional verification.

An experimental deployment on Ethereum proved the model: 30 venues competing in parallel per swap, atomic in-transaction fallback, and a settlement layer that enforces signed terms on every order. Pennysia v1, the Open Liquidity Network, generalizes that proof to every chain and every service class, a single interface for the world's liquidity.

This paper describes Pennysia v1, the Open Liquidity Network, as designed. Design decisions open at publication are noted where they occur; the Ethereum deployment documented in Appendix A demonstrates the core mechanisms in production.

1 Introduction

Crypto liquidity is fragmented across chains and venues with different access requirements [13]. The price available to a trader depends on where their assets are held and which providers they can access. Reaching a better quote elsewhere can require moving funds or opening another account, adding costs that reduce its advantage.

The **chain boundary** separates markets that maintain their own liquidity and settlement infrastructure. Execution quality varies by pair and order size, so a better quote may sit on a different network from the user's funds. Accessing it can involve bridge fees and settlement delays, as well as bridge risk [15]. The user must account for these costs and arrange execution across both networks before a price comparison becomes actionable.

The **access boundary** separates permissionless protocols from venues that require accounts or verification. These markets expose different liquidity and operate under different custody and settlement terms. Depth varies with market conditions and venue design [16], so either group may offer a better price for a given order. A trader seeking access to both must navigate their separate onboarding processes and trading interfaces.

For the user, this fragmentation compounds the practical work of self-custody. Direct control over assets [11] comes with responsibility for key management and transaction authorization, while gas and cross-chain transfers require additional coordination. Centralized exchanges consolidate much of this work within an account by holding assets on the user's behalf. Pennysia aims to provide comparable ease of access and execution speed while preserving user control over funds. Figure 1 illustrates the workflows the network seeks to simplify.

	On-ramp	*Wallet*	*Activity*	*Off-ramp*
DeFi	1. open a CEX or card-ramp account 2. pass KYC either way 3. buy the asset 4. wait out the withdrawal lock	5. install a wallet 6. write down the seed phrase 7. acquire the gas token 8. withdraw to the right chain	9. pick venue and chain 10. approve token (tx + gas) 11. swap (gas) 12. bridge if needed (gas + wait)	13. sell into a stablecoin 14. send back to a CEX 15. sell for fiat 16. bank withdrawal
CeFi	1. open account 2. pass KYC 3. deposit by card or bank	4. balance and keys held by the venue	5. trade on the venue's internal ledger	6. bank withdrawal

Figure 1: Illustrative fiat-in to fiat-out workflows. The self-custody example spans a ramp, wallet, and trading venue; the custodial example consolidates these steps within one platform. Pennysia aims to simplify the former through a unified account.

Intent-based execution [5] lets users authorize an outcome while an executor submits the transactions and pays gas. Embedded wallets [6] make user-controlled addresses accessible through familiar login methods. Together, these mechanisms support a common interface to a growing onchain market (Figure 2).

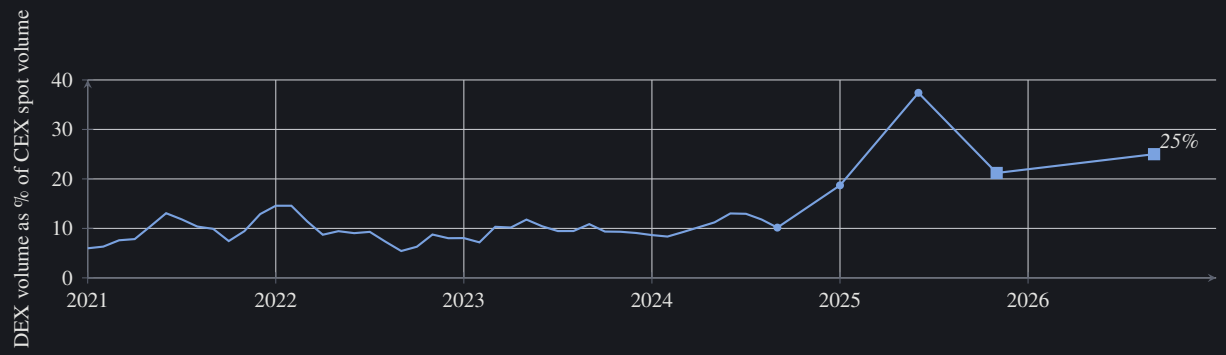


Figure 2: DEX spot volume as a share of CEX spot volume, 2021–2026. Line and circles use CoinGecko [7], monthly through Sep. 2024 and selected months thereafter. The square is The Block's reported ratio for Sep. 2026 [12]; the sources use different coverage and are not strictly comparable.

Pennysia v1, the Open Liquidity Network (OLN), extends the Ethereum prototype (Appendix A) across chain families and liquidity services. Its scope includes spot trading, lending, perpetuals, prediction markets, and vaults supplied by external providers. Pennysia provides the account, routing, and settlement coordination through which users access them, and a common integration through which providers compete for eligible orders.

2 Design Principles

The following principles guide implementation, including the choices that remain open.

- **Venue independence.** Providers integrate through a common executable-quote interface. Pennysia operates the access network and relies on external venues for liquidity and financial services.
- **User-controlled assets.** Pennysia holds no pooled user balances and cannot sign on a user's behalf. Users retain control of their wallets independently of the network.
- **Signature-based execution.** Users authorize the terms of an order. The network arranges transaction submission and gas payment on the required chains.
- **Execution-based ranking.** Quotes compete on output after costs, adjusted for fill probability. Listing is free, and venues cannot purchase ranking priority.
- **Optional verification.** Users can access permissionless venues without verification and opt into verification for eligible regulated services.
- **Disclosed costs.** Quotes include all execution costs. Any surplus fee applies only to output above the quoted amount. Access requires no native network token.

3 System Overview

The architecture separates the user interface and account from quote selection and execution (Figure 3).

Experience	one account, one balance, one ticket; chain & gas hidden, status to finality
Identity	login → non-custodial wallets (embedded or external); optional KYC
Quote market	parallel fanout → ranked net output; permissionless lane | regulated lane
Execution	signed order → best venue → verified fill; Settlement, intents, venue rails

Figure 3: Responsibilities of the four network layers.

Identity connects wallet access and verification to the account (Section 4). The quote market compares eligible providers (Section 5), then the execution layer settles the signed order using the appropriate rail (Section 6). The experience layer presents balances and order status throughout this process. Figure 4 shows how external participants connect to these layers.

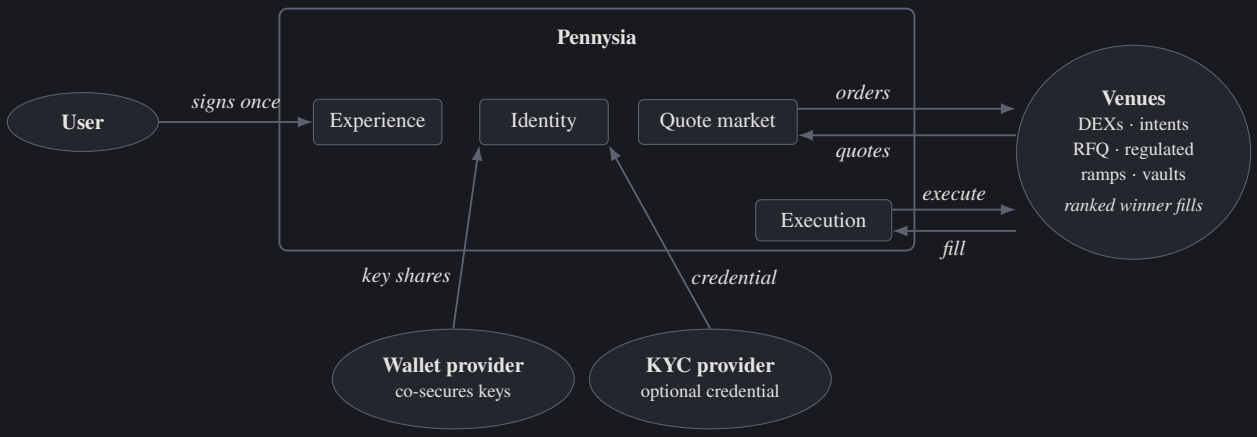


Figure 4: Participant interfaces. Wallet and KYC providers connect through Identity; venues exchange quotes and settlement information with the quote market and execution layers. The selected venue receives the signed order.

4 The Account

The account combines familiar login methods with user-controlled onchain wallets.

4.1 Onboarding and custody

A Web2 login (email, Google, or Apple) provisions **embedded wallets** for each supported chain family, initially EVM and Solana, without requiring a seed phrase or browser extension during signup. Assets reside at user-controlled onchain addresses, and moving them requires a wallet signature produced in the user's session. Table 1 defines each participant's authority.

Session verification authorizes account operations and quote assembly. It does not grant Pennysia signing authority.

4.2 Gas and wallet choice

Executors pay gas and include its cost in the quote (Section 6), so users can trade without acquiring a native gas token. Where a user must authorize an onchain action, such as an initial approval, the network sponsors or abstracts the gas.

External wallets, including extensions, WalletConnect connections, and hardware wallets, access the same quote market. The account provides a unified view and session for these wallets.

Table 1: Authority boundaries: what each actor can and cannot do.

Actor	Can do	Cannot do
User	Sign orders and transactions; export and self-manage keys	*none*
Pennysia	Verify sessions, assemble quotes, submit signed orders, display state	Move user funds; sign for the user; recover wallets
Wallet provider	Co-secure key material (device/enclave shares); assist user-driven recovery	Unilaterally spend

4.3 In and out

External ramp providers quote fiat-to-token and token-to-fiat conversions through the auction, with support for cards, bank transfers, Apple Pay, and Google Pay. These services sit in the regulated lane and require the applicable verification (Section 8). Pennysia coordinates access without holding a custodial balance.

The wallet provider handles recovery through re-authentication and restoration of key shares. Users can export and manage their keys independently of Pennysia.

Open design decision. The wallet implementation may use externally owned accounts with typed-data intents [2] or smart accounts [3, 4] supporting session keys and programmable recovery. This choice must be resolved in the detailed order specification (Appendix B).

5 The Quote Market

The quote market sends an order to eligible venues in parallel and compares their executable offers. Selection accounts for costs and the likelihood of settlement, both of which affect the user's realized output [14].

5.1 The auction

For user u, each venue v in the eligible set $E(u)$ returns a quote (y_v, c_v, p_v), where y_v is gross output, c_v is the all-in cost, and p_v is fill probability. Quotes must be executable commitments received within the market's response window τ. The simplified selection rule below expresses expected net output; the full ranking also accounts for settlement latency and per-quote eligibility requirements.

$$E(u) \;=\; \mathcal{V}_{\text{perm}} \;\cup\; \mathbf{1}[\text{verified}(u)] \cdot \mathcal{V}_{\text{reg}}, \qquad v^*(o) \;=\; \arg\max_{v \in E(u)} \underbrace{p_v \cdot (y_v - c_v)}_{\text{net output}} \tag{1}$$

The OLN extends the prototype's gas-aware quote comparison across chains and venue classes.

5.2 Two axes of liquidity

Providers differ by **permission**, which determines who may use a service, and **location**, which describes where its liquidity is held. These properties are independent; a venue can require verification while settling through an onchain contract.

Permission determines the user's *lane*. The **permissionless lane** includes open venues and the full prototype venue set. The **regulated lane** becomes available through optional verification, subject to venue eligibility (Section 8). Both participate in the same auction and ranking.

Liquidity location informs the execution rail (Section 6). Atomic and intent rails support onchain settlement, while venue rails coordinate delivery from offchain providers.

5.3 Venue and service classes

Figure 5 maps the landscape onto both axes.

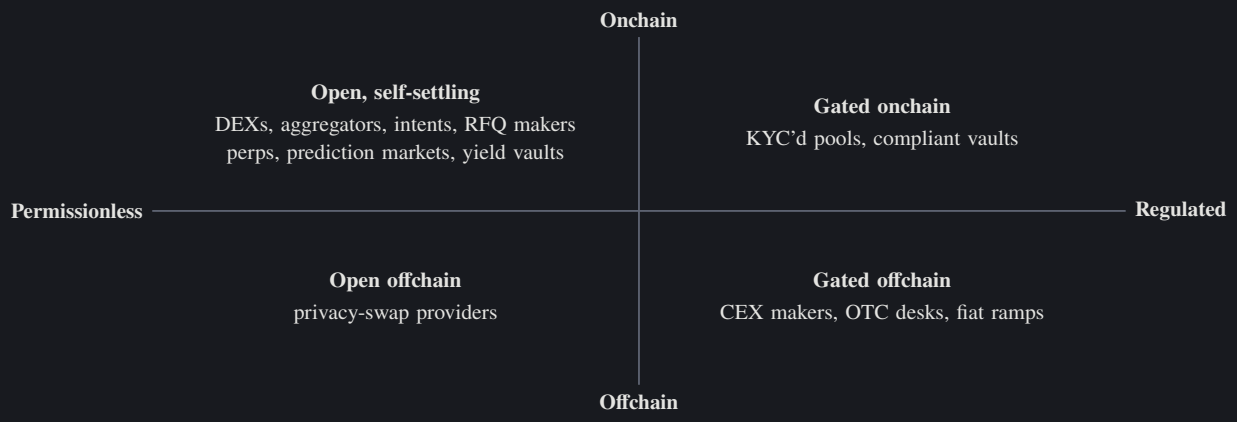


Figure 5: Provider classes by access requirements and liquidity location.

Services beyond swaps can participate when their adapters express executable quotes and verifiable settlement terms.

Permissionless integration is free and self-serve through the adapter interface. Regulated providers also complete onboarding for licensing, jurisdictional coverage, and settlement arrangements. Connected venues receive orders for which they are eligible.

5.4 Chainless quoting

The user selects the assets and desired outcome. The routing layer resolves the execution path across supported chains and venues, presenting net output and settlement time for review. Chain and rail details remain available for inspection.

6 Execution

The execution layer coordinates settlement within the terms authorized by the user.

6.1 Signature-first order flow

The user reviews the worst acceptable output (**WAP**, the minimum received after fees) and signs a typed-data order binding the input and output assets, output floor, deadline, and recipient (Appendix B). The network arranges the required transactions and gas payment.

6.2 The three rails

Orders settle over three rails (Figure 6).

Atomic routes. On a single chain, a Settlement-class router uses the signed allowance to pull the input and attempt ranked venues within one transaction. Execution stops at the first successful fill and reverts fully if all routes fail. This extends the prototype's production model to other EVM chains.

Intent rails. An intent settlement network matches the signed order with a filler that pays gas and delivers the specified output on the destination chain. Settlement is conditional on meeting the order's terms; an unfilled intent expires unpaid.

Venue rails. External venues use escrow contracts, where atomic exchange is possible [10], or deposit-address settlement with asynchronous delivery. The latter follows the prototype's privacy-swap integration and introduces counterparty exposure during settlement. Each quote discloses this model, and completion is reconciled against the signed terms (Section 7.4).

6.3 Guarantees and failure

A successful fill must deliver at least the signed minimum before the deadline. For order o signed by user u, the settlement condition is

$$\text{settled}(u,o) \implies \text{delivered}(u,o) \geq \text{WAP}(o) \ \wedge \ t \leq \text{deadline}(o) \tag{2}$$



Figure 6: Execution paths for a signed order. Each rail must reconcile its result against the authorized terms.

Failure handling depends on the rail. Atomic execution reverts, unfilled intents expire, and deposit-address settlement requires a refund if delivery fails. That refund depends on the venue's settlement arrangements and carries the counterparty risk described in Section 7.4. Table 2 summarizes the execution requirements.

Table 2: Execution requirements and supporting mechanisms.

Requirement	Mechanism
Worst-case output known at signing	WAP embedded in the order; enforced per rail
Gas abstraction	Executor pays gas, priced into quote; sponsored where unavoidable
Chain routing	Network resolves the execution path and destination
Failure handling	Revert or expiry; venue refunds subject to counterparty risk
Disclosed completion time	Rail-specific execution ETA; finality and slower settlement shown separately
One signature	Review → sign → status to finality

To match the responsiveness users expect from a centralized exchange, the design targets execution within a few seconds of receiving the signed order, down to sub-second execution where the venue and rail support it. Execution latency is measured from signed-order receipt to a verified fill, with blockchain finality tracked separately. Same-chain atomic fills depend on block inclusion, while cross-chain and asynchronous venue settlement may take longer. Quotes disclose the expected completion time for the selected rail, and order status distinguishes execution from final settlement.

Open design decision. The executor design remains open between per-chain Settlement contracts and a shared intent-native architecture. The choice determines deployment and audit scope.

7 Security Model

Security depends on venue controls, enforcement during execution, and protection of the user's signing keys.

7.1 Venue protections

Venue-specific protections, such as solver bonds or contract controls, continue to apply. Pennysia validates execution independently, while retaining exposure to the underlying venue's implementation and settlement model.

7.2 Enforced execution

Execution checks enforce the authorized terms and verify the resulting transfer.

- **Order validation.** Routes and quotes are checked against the signed payload before execution. Public-chain transaction reordering [8] requires validation of execution conditions.
- **Spending authority.** Atomic routes use a single Settlement approval, with transfers constrained by the authorized input and recipient.
- **Fill verification.** Settlement records require a match to the signed terms, including the output minimum and deadline. Atomic fallback follows the procedure in Section 6.2.

7.3 Account and key security

Table 1 separates account operations from signing authority. A compromised Pennysia session must not grant the server the ability to sign or move assets; wallet authorization remains the user's control over transfers.

7.4 Residual risks

The design retains the following dependencies and exposures.

- **Latency and filler availability.** Cross-chain settlement depends on filler liveness. Market movements or unavailable fillers can cause expiry and require a new quote. Multiple filler networks and alternative routes reduce this dependence where available.
- **Venue counterparty exposure.** Deposit-address settlement depends on the venue delivering output or honoring a refund. The quote discloses the settlement window and counterparty model.
- **Approval concentration.** A unified approval concentrates contract risk in Settlement, making its implementation and verification critical.
- **Wallet-provider dependence.** Embedded wallets rely on co-secured key material and provider-assisted recovery [9]. External wallets support users who prefer a different key-management model.

New contracts are intended to follow the prototype's verification process (Appendix A), with published source, invariant testing, and formal analysis where tractable. Deployment artifacts and verification results accompany each chain rollout. Common execution checks simplify integration but do not remove venue-specific risks.

8 Compliance Posture

Verification is optional for access to the network. It enables quotes from regulated venues where the user meets the provider's requirements. Eligibility is checked per venue and per quote, including jurisdictional restrictions, and is displayed before execution.

The goal is a reusable network-level verification credential that reduces repeated onboarding where venues accept it. Identity documents remain with the verification provider and the venue's own onboarding process; Pennysia coordinates credential status.

Open design decision. Venue acceptance will determine whether verification uses a shared attestation or separate onboarding with synchronized status.

Regulated coverage expands as venues are onboarded and may remain limited or unavailable in some jurisdictions. Venue-rail settlement carries the counterparty exposure described in Section 7.4. Pennysia enforces provider eligibility rules but does not determine users' legal or tax status.

9 Roadmap

Table 3 orders development by technical dependencies and defines a completion criterion for each phase.

Evaluation will track onboarding completion through the first settled trade, net fill quality against available alternatives, and venue participation in each lane. Execution reporting will measure median and 95th-percentile time from signed-order receipt to verified fill by rail, with time to finality reported separately. Settlement reporting will include fill-verification and failure rates.

Table 3: Proposed rollout and completion criteria.

Phase	Ships	Exit when
0. Prototype *(Ethereum mainnet)*	Parallel quoting and Settlement with atomic fallback; public contracts and verification reports	Quote selection and protected execution run on Ethereum mainnet
1. Account + multi-chain spot	Embedded EVM and Solana wallets, external-wallet support, and same-chain quoting across major EVM networks and Solana	A user signs in and swaps USDC without a seed phrase or native gas token
2. Chainless execution	Cross-chain intent execution, unified balances, and broader Settlement deployment	A cross-chain swap completes within the signed terms through the same account flow
3. Regulated lane	Verification, regulated venue and fiat-ramp integration, and jurisdiction-aware routing	An eligible user settles an order through a regulated venue
4. Network scale	Additional service classes and chain families, with a public order specification and integration APIs	Third-party fillers and integrators build on the network

A The Ethereum Prototype

This appendix records the experimental deployment on Ethereum Mainnet, live since August 28, 2026,[1] on which the OLN's design is based.

- **Parallel venue competition.** 30 venues quote per swap across four service classes (Table 4), reaching thousands of underlying liquidity sources. 12,000+ tokens listed.
- **Settlement contract.** Publicly available onchain router under BUSL-1.1 (`0x3Aad...795E`): single token pull, ordered venue fallback in one transaction, cross-chain intent routes, full revert on all-routes-fail.
- **Protections.** Signed output floor, gas-aware ranking, and unified Settlement approval, forming the basis of Section 7.
- **Fees live.** Surplus fee on outperformance (10% cap, hardcoded), flat per-order protocol fee, partner share on intent routes. Every fee folds into the signed quote.
- **Automated verification.** Slither (static analysis), Halmos (symbolic execution, 168 tests), Medusa (coverage-guided fuzzing, 223 tests), Certora (formal verification, 105 rules). The latter three use isolated component harnesses. Reports are public in the Settlement repository.
- **Defensive invariants.** Reentrancy guards via EIP-1153 [1]; venue/allowance-target may not equal the contract. Refunds are baseline-relative. Owner keys cannot raise the surplus cap.

Table 4: The prototype's 30-venue coverage by service class.

Class	Venues
DEXs & aggregators	onchain spot venues and meta-aggregators
Intent venues	same-chain and cross-chain intent networks
RFQ makers	quote-driven market makers
Privacy swaps	no-account, deposit-address providers

[1] Live at `https://app.pennysia.com`; launch announcement at `https://x.com/PennysiaLabs/status/2093384908354048180`; contracts and verification reports at `https://github.com/Pennysia/settlement`.

B Order Format

The following payload describes the proposed order fields, using EIP-712 typed data on EVM [2]. The final specification will define encoding and chain-specific signatures.

```
OLN-Order {
  version:      uint16       // format version
  signer:       address      // embedded or external wallet
  input: {
    chain:      chainId      // origin chain
    token:      address      // asset sold
    amount:     uint256      // exact input
  }
  output: {
    chain:      chainId      // destination chain
    token:      address      // asset bought
    minAmount:  uint256      // WAP floor - worst acceptable output
  }
  lanes:        bitmask      // PERMISSIONLESS | REGULATED
  recipient:    address      // output delivery address
  deadline:     timestamp    // order expiry
  nonce:        uint256      // replay protection + uniqueness
  signature:    bytes        // EIP-712 typed-data (EVM) / equiv. (Solana)
}
```

The output floor `minAmount` is the WAP. The `lanes` field limits eligible venues, with `REGULATED` requiring verification. Each rail uses these fields to validate delivery to the specified recipient on the destination chain before the deadline.

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